Tommaso Isidori

None

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Experimental Physicist & Data Acquisition Engineer.

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2. About Me

Ciao! I am Tommaso Isidori, born and raised in Siena (Italy), and later emigrated to follow my passions, as many of my generation did. This is even more common for the ones who followed my same career path and decided to pursue their love for Physics.
Convinced by my lab experiences during high school and thanks to the remarkable enthusiasm of my teachers I decided to start walking this rocky road during my Bachelor degree, which I pursued in my hometown, Siena. During my last year, an unexpected internship brought me to CERN for the first time in my life, introducing me to the High Energy Physics field, as well as a lot of colleagues and mentors that supported my momentum and guided me in my first steps in the world of physics.

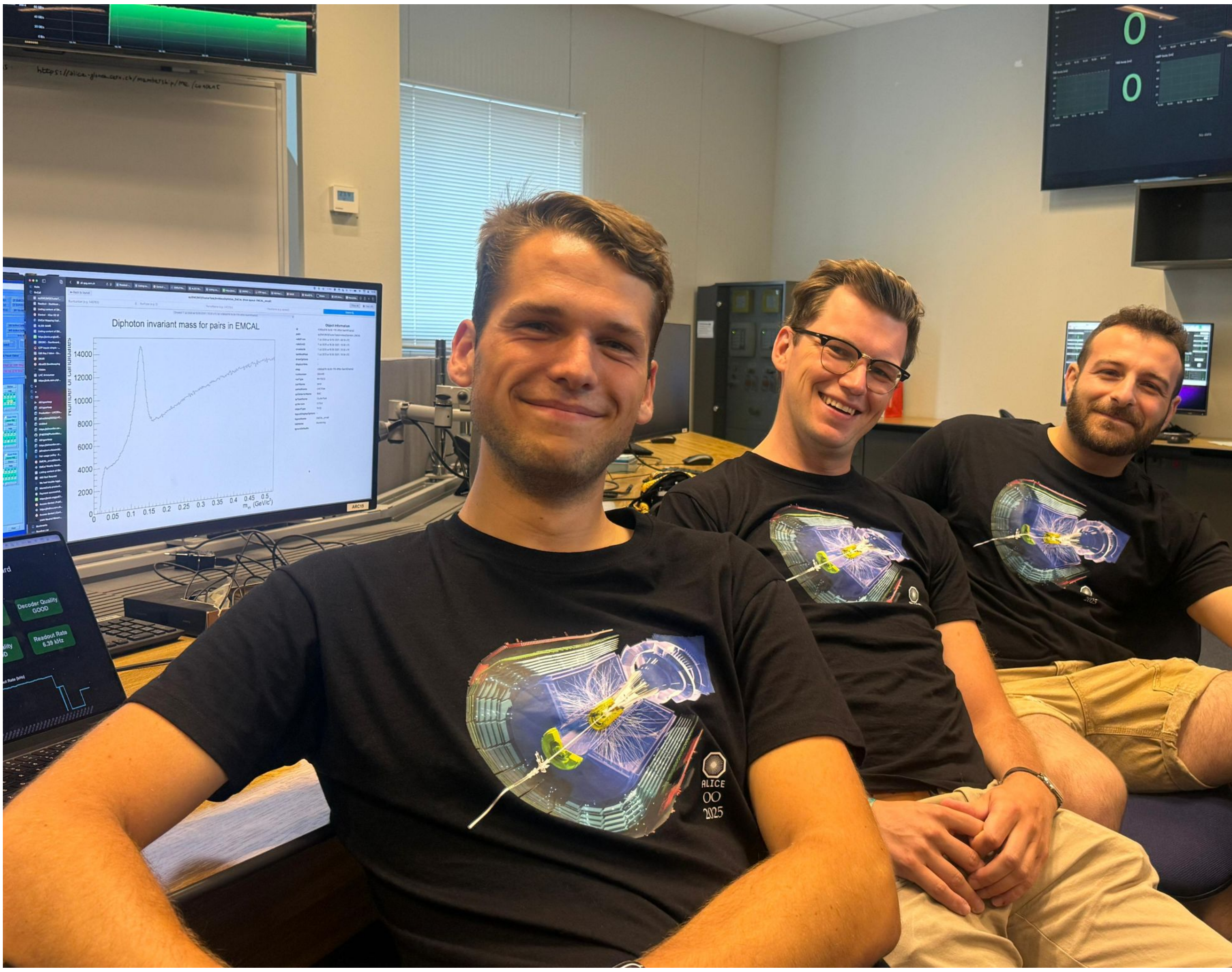
2.1 **My life as a Physicist**

I am an experimental nuclear physicist with a passion and experience in particle detectors. I am currently serving as a postdoctoral researcher at the University of Kansas, but I am based at CERN in Geneva (CH) to coordinate the test activities of my experiment.

Group photos from Test campaings (2022 - 2025) at CERN

I am deeply involved in the **ALICE FoCal experiment**, a forward calorimeter designed for the upgrade of the ALICE detector at the Large Hadron Collider (LHC). In this role, I serve as the **Test Beam Coordinator** and I am responsible for overseeing the FoCal laboratory. Since 2022 I directed the test, preparation, and optimization of the prototypes developed by our collaboration. ALICE FoCal is one of the two upgrade project of the ALICE experiment, at the LHC and more info can be found Here.

Recently, I led the operation of the **EMCal detector of ALICE**, holding the position of **Technical Coordinator** and **System Run Coordinator** during the LHC 2025 proton-proton campaign, as well as the scheduled special runs (Oxygen-proton, Oxygen-Oxygen, Neon-Neon). Although demanding under so many aspects, this parenthesis of my physics career represented an invaluable learning experience, and provided me with a profound understanding of the operation of a running detector, and the privilege of working side by side with some of the best colleagues I met along my route.



N.Strangmann (left), F.Jonas (center), and I during the 2025 LHC light ion data acquisition period

Beam monitoring of a high-rate linac for radiotherapy (Dublin, Ireland)

As part of my spin-off projects, I also contribute to detector **Research and Development (R&D)** efforts for calorimetry and fast timing for the future Electron-Ion Collider (EIC). My involvement in this is a natural continuation of my studies on radiation-hard solid-state timing detectors for HEP and beam diagnostic pursued during my Master and PhD reasearch. I had the chance to test novel solid-state detectors specialized in high-rate measurements for beam monitoring, with the use of particle beam operated at high intensities in medical and research facilities.

Previously, I completed my **Ph.D. at the University of Kansas** under the supervision of Professor Christophe Royon. My doctoral studies were centered on particle physics, with a focus on the design, testing, and characterization of particle detectors for High Energy Physics experiments. My expertise spans various detector types, including timing, solid-state, and gas detectors. During my Ph.D., I contributed to two particle physics collaborations: the **Compact Muon Solenoid (CMS)** and the **TOTEM/PPS** collaborations. My thesis topics also included additional detector R&D projects, such as the design of a **NASA cubesat detector** for particle identification and the employment of **fast Si detectors** for single particle identification in high-rate medical facilities.

2.2 **What else?**

3. Tommaso Isidori

Experimental Physicist & Data Acquisition Engineer.

- Download short CV
- Download extended CV

4. CV Preview

TOMMASO ISIDORI

Experimental Physicist & Data Acquisition Engineer

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EXPERIENCE

Postdoctoral Researcher

CERN, ALICE Experiment

2021 – Present | CERN, Geneva (CH)

- Coordinate test campaigns, managing hardware, data collection, and analysis across international teams.
- Develop and maintain DAQ, readout, and control systems (C++, Python, Bash) and embedded monitoring tools.
- Support system operations, overseeing software, DAQ configuration, and on-call team training.
- Lead R&D on fast timing detectors, developing DAQ and signal analysis frameworks and mentoring students.

Ph.D. Researcher

University of Kansas (KU)

2017 – 2022 | Lawrence, KS, USA

- Designed and characterized silicon detectors and front-end electronics for space, medical, and HEP applications.
- Led test beams and lab setups for fast silicon detector characterization at CERN and NASA.
- Conducted high-intensity radiotherapy detector studies, achieving single-particle resolution for dosimetry.

Internship at INFN Pisa

Universita degli studi di Pisa

2016 – 2017 | CERN, Geneva (CH)

- Designed, tested, and integrated high-speed solid-state sensors and electronics under high-radiation conditions.
- Led validation campaigns, developing DAQ software and performing system installation, calibration, and optimization.

Summer student at the TOTEM experiment

Universita degli studi di Siena

2013 | CERN, Geneva (CH)

- Developed and tested gas-based detectors, performing lab measurements and system characterization.
- Designed feedback systems for low-current monitoring, collaborating with electronics engineers on signal reliability.

ROLES AND AWARDS

Prototype performance coordinator of the ALICE FoCal project
2021 – current

Technical Coordinator and System Run Coordinator of the ALICE EMCal experiment
2024 – 2025

Responsible of laboratories at CERN
2021 – current

Awardee of the KU Graduate Fellowship Based on Exceptional Qualifications
2017 – 2018

STRENGTHS

Hard-working | Team work | Embedded Systems | Sensors & Detectors | Statistical Analysis | Coordination & Management

EDUCATION

Ph.D. in Physics

The University of Kansas (KU)

2017 - 2022 | Lawrence, KS, USA

Thesis title: *New Generation Fast Silicon Detectors for Timing and Particle Identification: High Energy Physics and Applications*

M.Sc. in Physics, Fundamental Interaction

Universitá degli studi di Pisa

2014 - 2017 | Pisa, Italy

Thesis Title: *Diamond Detectors for Proton Time of Flight in CT-PPS and TOTEM Experiments*

B.Sc. in Physics

Universitá degli studi di Siena

2010 - 2014 | Siena, Italy

Thesis Title: *Characterization and Optimization of Gas Detectors*